

Phase 9 CVRP Suite Report

Overview

- Condition count: 6.
- Best held-out condition: phase9_closeout_baseline_clarke_wright_savings.

Condition Summary

Condition	Mode	Held-out Feasibility	Held-out Penalized Gap	Held-out Feasible Gap	Held-out Runtime (ms)	Accepted Epochs	Fallback Epochs	Generation Errors	Mean Novelty	Mean Complexity
phase9_closeout_baseline_nearest_neighbor_constructive	baseline_nearest_neighbor_constructive	1.0	0.273435	0.273435	42.3532	0	0	0	0.0	0.0
phase9_closeout_baseline_clarke_wright_savings	baseline_clarke_wright_savings	1.0	0.073766	0.073766	108.75142	0	0	0	0.0	0.0
phase9_closeout_baseline_regret_insertion_local_search	baseline_regret_insertion_local_search	1.0	0.466391	0.466391	1434.62148	0	0	0	0.0	0.0
phase9_closeout_direct_generate_plus_one_repair	direct_generate_plus_one_repair	0.0	3.0	n/a	198.82434	1	0	0	0.969708	15.85
phase9_closeout_budget_matched_no_replay	budget_matched_no_replay	0.0	3.0	n/a	239.00458	2	0	0	0.965757	12.37
phase9_closeout_replay_solver_evolution	replay_solver_evolution	1.0	0.273435	0.273435	139.51872	0	0	0	0.0	0.0

Condition Notes

phase9_closeout_baseline_nearest_neighbor_constructive

- Execution mode: baseline_nearest_neighbor_constructive.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.337704.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.273435.
- Held-out runtime ms: 42.3532.
- Accepted epoch count: 0.
- Fallback epoch count: 0.

- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.002777.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.408032, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.364218, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.317111, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.145642, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.132172, errors=[]

phase9_closeout_baseline_clarke_wright_savings

- Execution mode: baseline_clarke_wright_savings.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.064783.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.073766.
- Held-out runtime ms: 108.75142.
- Accepted epoch count: 0.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.007791.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.108521, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.099285, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.061249, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.057708, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.042065, errors=[]

phase9_closeout_baseline_regret_insertion_local_search

- Execution mode: baseline_regret_insertion_local_search.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.451049.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.466391.
- Held-out runtime ms: 1434.62148.
- Accepted epoch count: 0.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.086899.
- Worst held-out instances:

- X-n275-k28: feasible=True, penalized gap=0.746105, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.650864, errors=[]
- X-n247-k50: feasible=True, penalized gap=0.395235, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.390341, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.149411, errors=[]

phase9_closeout_direct_generate_plus_one_repair

- Execution mode: direct_generate_plus_one_repair.
- Train feasibility rate: 0.0.
- Train penalized gap: 3.0.
- Held-out feasibility rate: 0.0.
- Held-out penalized gap: 3.0.
- Held-out runtime ms: 198.82434.
- Accepted epoch count: 1.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.0.
- Worst held-out instances:
 - X-n176-k26: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
 - X-n190-k8: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
 - X-n223-k34: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
 - X-n247-k50: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
 - X-n275-k28: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']

phase9_closeout_budget_matched_no_replay

- Execution mode: budget_matched_no_replay.
- Train feasibility rate: 0.0.
- Train penalized gap: 3.0.
- Held-out feasibility rate: 0.0.
- Held-out penalized gap: 3.0.
- Held-out runtime ms: 239.00458.
- Accepted epoch count: 2.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.0.
- Worst held-out instances:
 - X-n176-k26: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
 - X-n190-k8: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
 - X-n223-k34: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']

- X-n247-k50: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
- X-n275-k28: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']

phase9_closeout_replay_solver_evolution

- Execution mode: replay_solver_evolution.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.337704.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.273435.
- Held-out runtime ms: 139.51872.
- Accepted epoch count: 0.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.002777.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.408032, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.364218, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.317111, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.145642, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.132172, errors=[]

Judge Note

Phase-9 CVRP Closeout Suite Summary (Conservative)

Aspect	Summary
Direct Synthesis	direct_generate_plus_one_repair failed completely on feasibility (0% feasible), very high penalized gap (3.0).
Budget-Matched Independent Search	budget_matched_no_replay also failed feasibility (0%) with large gap (3.0).
Replay-Aware Iterative Search	replay_solver_evolution achieved perfect feasibility (100%) but with a moderate feasible gap (0.273), not better than best baseline.
Fixed-Baseline Context	Baselines clarke_wright_savings, nearest_neighbor_constructive, and regret_insertion_local_search all reached 100% feasibility.
Best Baseline	clarke_wright_savings was best on heldout feasible gap (0.074), with moderate runtime (109 ms), and good robustness.
Feasibility	Baselines: 100%, Replay: 100%, Direct/Budget-matched No-Replay: 0%.
Objective Gap	Baselines: best gap ~0.07 (Clarke-Wright), Replay gap ~0.27, Direct and No-Replay gaps: 3.0 (failed).
Runtime	Baselines ranged from ~42 ms to 1435 ms; Replay runtime ~140 ms; Direct and No-Replay higher (~200-240 ms).
Robustness	Baselines robustness dispersion low; Replay similar to nearest_neighbor; direct/budget methods no robustness as feasibility failed.

Conclusions

- ****No learned condition (direct synthesis, budget-matched no-replay, replay-evolution) clearly beat the best fixed baseline (clarke_wright_savings) in feasibility or objective gap.****
- ****Replay iteration (replay_solver_evolution) did not outperform the budget-matched no-replay control in objective gap or feasibility; it matched baseline feasibility but had a significantly worse gap than best baseline.****
- ****Direct synthesis and budget-matched no-replay controls both failed feasibility entirely despite moderate runtimes.****

****In brief:**** Baseline heuristic solvers remain superior in feasibility, quality, and robustness. Learned methods, including replay-aware search, do not surpass these fixed baselines under the current phase-9 CVRP closeout evaluation.